4 Conversation as mechanism: emergence in creative groups

Keith Sawyer

Most explanations are causal explanations: "I ate a snack because I was hungry." This is a very simple causal explanation: the event of "being hungry" caused the event of "eating a snack." However, this statement, connecting two states at two successive moments in time, does not provide the full causal story connecting these two events. How, exactly, did "being hungry" cause "eating a snack"? A mechanist would argue that the complete explanation would have to describe the underlying mental state of "being hungry" and how this mental state then caused the neurons in the brain to make the body move into the kitchen and prepare the snack. This would be a very complicated story involving millions of neurons, as well as the relations between the stomach, the brain and the body.

This simple example demonstrates a general property of mechanistic explanations: they commonly describe processes in very complex systems that underlie what, on the surface, appears to be a simple causal relationship. Advocates of mechanistic explanation are most likely to be found in the philosophy of biology and in the philosophy of the social sciences, because both biology and the social sciences are centrally concerned with complex systems. A mechanistic explanation of an event (getting a snack) traces the causal processes leading up to that event by describing the components of a complex system and their interactions (the brain, the stomach sensors that detect hunger). They are *generative* explanations; they explain how the operations of the system generate the observed phenomena.

Many mechanists go beyond this definition to make two additional claims that I reject in this chapter. First, many mechanists define mechanistic explanation in opposition to deductive-nomological (DN) or covering law approaches. The DN approach explains an event by identifying initial conditions and a series of lawful generalizations that show how the event can be deduced from the initial conditions. An example of a covering law that accounts for my simple example, how to explain the act of eating a snack, would be the lawful generalization "When a

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person is hungry, they are likely to eat" combined with the initial condition, "I am hungry," thus allowing the observer to deduce that I will prepare some food. I argue that the opposition of mechanistic explanation to the DN approach has been over-emphasized and is not necessary to the definition of mechanism. After all, even after the complete neurological account of hunger and snack-preparing has been developed, the simple regularity "hunger causes eating" still holds true (and is no doubt more useful in everyday life than the elaborately detailed account involving millions of neurons). Here I recommend Pierre Demeulenare's chapter in this volume, Chapter 9, which explores the notion of regularities versus special cases and argues that contra Peter Hedström, mechanistic and covering-law explanations are compatible; and in fact, they are more interrelated than most of us believe.

Second, many mechanists make the additional claim that a mechanistic explanation must be a lower-level one; a higher-level phenomenon is said to be explained when the lower-level system that gave rise to the phenomenon is sufficiently described: the components of the system, and their interactions. In this view, "hunger" is a mental state that must be explained in terms of the neuronal configuration of the brain that corresponds to that state. This form of mechanism is equivalent to methodological individualism in the social sciences, and to physicalist reductionism in the philosophy of science more generally. To perhaps take my hunger example a bit too far, imagine the twenty-person kitchen staff in a three-star restaurant; the explanation of how the kitchen works to generate hundreds of multi-course meals in one evening is a mechanistic account in terms of individual actions and interactions. The components of that system are individuals, just as the components of the brain are neurons. In both cases, the mechanistic explanation works at a lower level of analysis, describing system components and their interactions.

In this volume, Sperber's accounts of mental states (Chapter 3), and Elster's account of emotions (Chapter 2), are mechanistic explanations, and yet they do not refer to neurons. They present accounts of psychological mechanisms, not neurological mechanisms. By analogy, just as psychological mechanistic explanations can be constructed that do not have neurons as their components, social mechanist explanations can be constructed that do not have individuals as their components. Thus, contrary to most social mechanists, I argue that social mechanism is not definitionally identical with methodological individualism (just as psychological mechanism is not identical to neuroscience). Mechanisms exist at many levels of analysis. A sociological explanation could be a causally mechanist explanation even if it does not concern properties of

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individuals. One could provide mechanistic explanations of large-scale social systems in which the components are smaller-scale social units (for Auguste Comte, it was the family, for example).

In my empirical studies of creative groups, including jazz and theater ensembles, I explain their performances by providing mechanistic accounts in which communicative interactions are the fundamental unit of the system. This approach is grounded in a long tradition of sociological study of interaction – including symbolic interactionism, ethnomethodology and conversation analysis – that goes back to at least Georg Simmel. In empirical practice, a focus on interaction has led to a focus on social mechanisms; social psychologists in general have focused on the interactional mechanisms that give rise to the emergence of group properties. The key mechanisms in these groups involve symbolic communication.

Those social mechanists who propose reductionist and individualist accounts of emergent social phenomena generally incorporate only simplistic accounts of human communication. In this chapter, I argue that social mechanistic explanations are strengthened when they incorporate the sophisticated symbolic aspects of communication. However, once symbolic communication is incorporated into a social mechanistic explanation, it becomes exceedingly difficult to formulate that explanation in terms of methodological individualism.

Improvisational encounters

A successful improvised performance is an emergent phenomenon, one that can be observed as it unfolds on stage. After years observing these creative groups – jazz and improvisational theater ensembles – I developed a strong intuition that the whole was greater than the sum of the parts. For example, the performance that emerges is not predictable, even knowing a great deal about the members of the ensemble. The performance that emerges cannot be reduced to the mental states and intentions of the individual performers. And the performance that emerges is novel – a unique creation. These three features – unpredictability, irreducibility and novelty – are classic characteristics of emergent phenomena (Sawyer 2005).

In an improvisational theater performance, how does the interactional frame emerge – the characters, motivations, relationships, and plot events and sequence? No single participant creates the frame; it emerges from the give and take of conversation. The frame is constructed turn by turn; one person proposes a new development for the frame, and others respond by modifying or embellishing that proposal.

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Each new proposal for a development in the frame is the creative inspiration of one person, but that proposal does not become a part of the frame until it is evaluated by the others. In the subsequent flow of dialog, the group collaborates to determine whether to accept the proposal, how to weave that proposal into the frame that has already been established, and then how to further elaborate on it.

Example 1 is the first few seconds of dialog from a scene that the actors knew would last about five minutes. The audience was asked to suggest a proverb, and the suggestion given was "Don't look a gift horse in the mouth."

Example 1. Lights up. Dave is at stage right, Ellen at stage left. Dave begins gesturing to his right, talking to himself.

1	Dave	All the little glass figurines in my menagerie, The store of my dreams. Hundreds of thousands everywhere!	Turns around to admire.
2	Ellen		Slowly walks toward Dave.
3	Dave		Turns and notices Ellen.
		Yes, can I help you?	
4	Ellen	Um, I'm looking for uh, uh, a present.	Ellen is looking down like a child, with her fingers in her mouth.
5	Dave	A gift?	
6	Ellen	Yeah.	
7	Dave	I have a little donkey.	Dave mimes the action of handing Ellen a donkey from the shelf.
8	Ellen	Ah, that's – I was looking for something a little bit bigger …	
9	Dave	Oh.	Returns item to shelf.
10	Ellen	It's for my Dad.	

By turn 10, elements of the frame are starting to emerge. We know that Dave is a storekeeper, and Ellen is a young girl. We know that Ellen is buying a present for her Dad and, because she is so young, probably needs help from the storekeeper. These dramatic elements have emerged from the creative contributions of both actors. Although each turn's incremental contributions to the frame can be identified, none of these turns fully determines the subsequent dialog, and the emergent dramatic frame is not chosen, intended or imposed by either of

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the actors. (Also note that the dialog is not obviously derived from the audience's suggestion, either, although the actors will later integrate it with the emerging frame.)

The emergence of the frame cannot be reduced to actor's intentions in individual turns, because in many cases an actor cannot know the meaning of her own turn until the other actors have responded. In turn 2, when Ellen walks toward Dave, her action has many potential meanings; for example, she could be a co-worker, arriving late to work. Her action does not carry the meaning "A customer entering the store" until after Dave's query in turn 3. In improvised dialogs, many actions do not receive their full meaning until after the act has occurred; the complete meaning of a turn is dependent on the flow of the subsequent dialog. This sort of retrospective interpretation is quite common in improvised dialog, and it is one reason that the emergent frame is analytically irreducible to the intentions or actions of participants in individual turns of dialog.

Improvisational encounters are different from the social phenomena that are usually studied by social mechanists. Social mechanists typically focus on middle-range phenomena, such as unemployment in Stockholm (Hedström 2005). The most immediate contrast is that the number of agents is quite small: in Example 1, there are only two agents. In contrast, many mechanistic explanations – particularly those using the computational simulation technique known as agent-based modeling – have hundreds or thousands of autonomous agents (e.g. Epstein 2006; Hedström 2005). There are three additional differences that lead me to argue that even though these social systems have as few as two agents, they nonetheless demonstrate the usual properties of emergence – including unpredictability, irreducibility and novelty. These three are:

1. a broad range of possible actions;
2. retroactive interpretation;
3. downward causation.

A broad range of possible action

At each point in the improvisation, an actor can choose from a wide range of moves to propel the dramatic frame forward. This unpredictability results in dialog that, at each turn, has a combinatorial complexity: a large number of next turns is possible, and each one of those turns could result in the subsequent flow of the dialog going in a radically different direction. This results in expanding combinatorics such as those in a chess game. Such moment-to-moment combinatorics often result in analytically irreducible phenomena.

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Retroactive interpretation

It is often impossible to determine the meaning of a turn at the moment that the turn occurs. In Example 1, when Ellen walks toward Dave, her action has several potential meanings, and these ambiguities are not resolved until the subsequent turn of dialog. Before the subsequent flow of the interaction attributes a meaning to it, each of the two actors may have had a different interpretation of what the walking meant; it is not until later that these conflicting potential interpretations are resolved. This unfolding intersubjectivity results in collaboratively emergent phenomena that cannot be reduced to an analysis of the component individuals.

Downward causation

An individualist account of Example 1 would analyze each turn in terms of the causal influence of the prior turn of dialog, combined with the intentions and mental models held by the speaker, without any appeal to an analytically distinct social level of analysis. But with improvised dialogs, it is difficult to account for an individual's actions without making explicit reference to the independent causal force exerted by the emergent interactional frame.

The collaborative emergence of frames has been studied by several researchers in interactional sociolinguistics and conversation analysis, including Deborah Tannen, Alessandro Duranti and Charles Goodwin (Duranti and Goodwin 1992; Tannen 1993). These researchers have generally avoided arguing that the emergent frame is a real social phenomenon with autonomous social properties (Sawyer 2003b). Rather, the frame is considered to exist only to the extent that it is "demonstrably relevant" (Schegloff 1992) to participants – a classic interpretivist stance, one that rejects social realism.

In contrast, I argue that in improvised dialogs, the emergent interactional frame must analytically be considered to have its own causal force. Most sociologists generally believe that the causal impact of macro-social forces must be mediated through an individual's perception of them. However, in improvised dialog, people act without conscious awareness or reflection. If causal relations of constraint and enablement can be empirically identified in improvised dialogs, this would be evidence in support of the more general sociological claim that social emergents can have causal effects on individuals. I have identified such relations in several extended analyses of long improvised dialogs (Sawyer 2003b).

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Improvisation as explained by methodological individualism

An individualist would hold that the interactional frame is nothing more than representations of it in the minds of individual participants; there is no higher level of analysis at which the frame emerges. The methodological individualist would explain improvised dialogs in terms of the mental representations of the frame held by the participants. The frame is not an autonomous social reality and it has no causal power.

Hedström and Swedberg's (1998) introductory chapter explicitly noted that "the principle of methodological individualism is intimately linked with the core idea of the mechanism approach" (p. 12), one of their four principles was "the general reductionist strategy in science" (p. 25), and both their examples and their typology of social mechanisms were methodologically individualist. They closed by implicitly indicating their opposition to social realism: "there exist no such things as 'macro-level mechanisms'." Thus social mechanists implicitly and explicitly deny a form of social realism that holds that emergent social properties can have downward causal powers (as advocated by the emergentists Archer 1995 and Bhaskar 1975/1997).

I hold that laws are compatible with mechanisms; mechanisms "explain" laws (Beed and Beed 2000; Bunge 2004; Elster 1998). However, it is possible that social laws may exist that are difficult to explain by reduction to micro mechanisms; if so, the scope of methodological individualism would be limited. In an earlier work (Sawyer 2005) I provided an account of emergence that I called *nonreductive individualism (NRI*) that shows how this could be so. I argue that some emergent social properties may be real, and may have autonomous causal powers, just like real properties at any other level of analysis. To the extent that social properties are real, social mechanist explanation may be limited to the explanation of individual cases that do not generalize widely, resulting in a case study approach rather than a science of generalizable laws and theories.

In developing NRI, I was heavily influenced by arguments surrounding the mind–brain relation; and many philosophers believe that these arguments can be generalized to apply to any hierarchically ordered sets of properties (Fodor 1989; Humphreys 1997: 3; Jackson and Pettit 1992: 107; Kincaid 1997: 76; Yablo 1992: 247, n. 5). NRI holds to a form of *property dualism* in which social properties may be irreducible to individual properties, even though social entities consist of nothing more than mechanisms composed of individuals.

The argument uses notions of supervenience, multiple realizability, and wild disjunction to argue that the relation between emergent social

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properties and their realizing mechanisms is one of token identity, but not of type identity. In other words, on any token occasion, a social property must be realized by a mechanism involving its component individuals. However, on different occasions the same social property might be realized by different mechanisms (multiple realizability), and those different mechanisms might not be similar in any sociologically meaningful way (wild disjunction).

It is possible that some social properties that participate in causal laws are multiply realized in wildly disjunctive systemic mechanisms. If so, that social property is real and it has autonomous causal power (Sawyer 2003c). Thus like Fodor (1974) in the philosophy of mind, and Kincaid (1997) in the philosophy of social science, in such cases I argue that causal explanation need not cite underlying mechanisms. For example, the social property "competitive team sport" can participate in causal laws, and can play an important role in sociological explanation. Such explanations are quite standard in science more generally; each science standardly holds that its types and properties are real and that its properties participate in causal laws. Pressure laws hold regardless of what particles make up the gas, and regardless of the details of their relations and interactions. Airfoil laws hold with cloth sails and steel airplane wings, in varying temperature and humidity.

The methodological individualist's usual response to multiple realizability arguments is to claim that in such cases the many realizations of a given social property are lawfully and meaningfully related, so that multiple realization is a trivial and ignorable detail (e.g. Macdonald and Pettit 1981; Pettit, personal communication, 2002). A methodological individualist would have to argue that if a given social property is realized on different occasions by different mechanisms, those different mechanisms would nonetheless be quite similar and describable in basically the same explanatory framework. The methodological individualist might argue that on all occasions when a group has the property "competitive team sport," the individual members have similar intentional states (such as "belief") and the members are organized according to similar interpersonal bonds (such as "solidarity").

In the case of improvised dialogs, it is particularly difficult to make this argument.

An expanded conception of social mechanism

In improvised dialogs, there is always a continuing dialectic: social emergence, where individuals are co-creating and co-maintaining emergent interactional frames; and downward causation from those emergent frames. During conversational encounters, interactional

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frames emerge, and these are collective social facts that can be characterized independently of an individual's interpretations of them. Once a frame has emerged, it constrains the possibilities for action. Although the frame is created by participating individuals through their collective action, it is analytically independent of those individuals, and it has causal power over those individuals. I refer to this process as *collaborative emergence* (Sawyer 2003a) to distinguish it from models of emergence that fail to adequately theorize interactional processes and emergence mechanisms. To account for these processes, sociologists must develop an account of the mechanisms of collaborative emergence that lead to these emergents.

When examining the extensive theoretical literature on emergence, in both philosophy and sociology, one finds a great deal of confusion, differences in terminology, and disagreement about the exact nature of emergent phenomena. For example, a key question remains unresolved: can emergent phenomena have autonomous downward causal power over the participating individuals? In improvisational theater, my empirical observations seemed to demonstrate "downward causation" – the emergent dramatic frame had constraining and enabling effects on the individual actors. I identified downward causal effects of emergent phenomena, through detailed interaction analyses of specific theater performances (Sawyer 2003b). As a consequence of these empirical observations, I developed the above theory of nonreductive individualism.

Emergence theories have existed long before the recent turn to mechanistic theories. Nonetheless, emergence and mechanism are quite compatible – in that both approaches emphasize processes and interactions among components in complex systems. Mechanistic approaches began to emerge in the 1990s, both in the philosophy of biology (Bechtel 2001; Bechtel and Richardson 1993; Craver 2001, 2002; Glennan 1996; Machamer *et al*. 2000) and in the philosophy of the social sciences (Elster 1989; Hedström and Swedberg 1998; Little 1998; Stinchcombe 1991).

Most social mechanists reject covering-law explanations of social processes and phenomena (cf. Markovsky 1997; Turner 1993). Rather than explanation in terms of laws and regularities, mechanists argue that sociology should provide explanations by postulating the processes constituted by the operation of mechanisms that generate the observed regularities. Such explanation is provided by the specification of often unobservable causal mechanisms, and the identification of the processes in which they are embedded (Hedström and Swedberg 1998). As such, mechanistic approaches implicitly assume a realist perspective

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and reject empiricism (Aronson *et al*. 1995; Bhaskar 1975/1997; Layder 1990).

I believe that sociology has much to gain by a turn toward mechanism. In particular, I believe that an adequate account of emergence requires a mechanistic explanation. And as an example, I argue that mechanistic approaches are required to adequately explain improvised dialogs. I furthermore argue that these creative conversational encounters are at the core of all socially emergent phenomena, and that a mechanistic account of creative conversation is a necessary component of any account of social life. However, the appropriate mechanistic account would not be strictly individualist; it would incorporate individuals, symbolic interactions and emergent properties of the interactional frame.

All mechanistic accounts are necessarily reductionist in the sense that they reject "strong emergence" – a form of emergence in which emergent properties have autonomous causal power over the components of the system. Only within a covering-law framework can higher-level properties have causal power over lower-level properties. There can be no mechanistic downward causation within a single token system; downward causation can only exist as a lawful generalization across multiple cases (which are multiply realized). Yet, I argue that mechanists go too far in claiming that there can be no strong emergence; it is true that it cannot exist within mechanistic explanation, but it can hold for covering-law explanations.

Conversations are the mechanisms of social emergence

Sociologists who study emergence rarely incorporate the study of symbolic interaction; and sociologists who study symbolic interaction rarely consider emergence. Our goal should be to develop a theoretical framework that incorporates both symbolic interaction and emergence. Toward that goal, I begin this final section by briefly summarizing how these topics have been explained by two prominent paradigms in sociological theory over the past century: the structure paradigm, and the interaction paradigm (this is a summary of the more extensive account that appears in Sawyer 2005). The structure paradigm focuses on the relations between two distinct levels of analysis: the *individual* and the *social*. Most social mechanist accounts are of this type, in that they explore relations between two distinct levels of analysis, the individual and the social. The interaction paradigm is distinguished by its focus on an additional, intermediate level of interaction, symbolic action in

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communicative episodes. I argue that neither of these two theoretical strands can adequately account for conversation and emergence. The solution is to take a mechanistic approach, one that can unify the best elements of both the structure and the interaction paradigms.

The structure paradigm

Social mechanists generally work within a long tradition in sociology that I call the structure paradigm (Sawyer 2005). The defining feature of this paradigm is its focus on the relations between two levels of analysis, the individual and the social. Traditionally, sociologists within the structure paradigm have fallen into two opposed camps: collectivists and individualists. Collectivists emphasize the ontological autonomy of the social level, and its causal power over individuals. Individualists argue that collective phenomena are epiphenomenal, and that all social forces must operate through the perceptions of them by individuals. What unifies these opposed camps is that both assume that conversation is epiphenomenal – it has no causal consequences, either for emergent macro phenomena or for individuals. Instead, the ultimate causal forces in social life are either institutions, networks and group properties (for the collectivist), or rational actions taken in the context of pairwise game-like encounters (for the individualist).

More recently, a third camp has emerged that blends elements of both; I call these hybrid theories. The most advanced hybrid theories are of the micro–macro link (e.g. Alexander *et al*. 1987; Knorr-Cetina and Cicourel 1981) and of the structure–agency link (Archer 1995; Giddens 1984). Although a theory of emergence must be a theory that relates the individual and the collective levels, even these advanced versions fail to explain emergence, because they have inadequate theories of the interactional mechanisms that connect the two levels of analysis. Ultimately, the structure paradigm cannot explain social emergence, because it does not incorporate theories of process, mechanism and interaction.

The interaction paradigm

The structure paradigm fails to account for emergence because it does not theorize the interactions and processes within a system's mechanisms. Thus it does not theorize the critical mediating link between the individual and the social levels of analysis: interactions between individuals. In the 1960s and 1970s, important alternatives to the structure paradigm began to emerge; I group them together using the term *interaction paradigm*. The interaction paradigm represented a fundamental

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break; it focused on the processes and mechanisms of interaction neglected by the structure paradigm. Although this was a necessary step forward, ultimately I conclude that the interaction paradigm fails to account for emergence, although for very different reasons than the structure paradigm.

The defining feature of the interaction paradigm is the belief that properties of interaction are not derivable from the individual actions or agency of the members of the group, nor can they be derived from social structure. Interaction is an ontologically distinct level of analysis. There are fundamental properties and laws of interaction – based, for example, on semiotics, cybernetics or communication theory – that are not reducible either to individual properties or structural characteristics. Because interaction is not reducible to individuals or to structure, it is an autonomous level of analysis. The claim that interaction is a level of analysis with ontological status implies that it possesses properties that participate in causal relations, with causal effects on both structure and individuals.

Interaction is at the center of an old sociological tradition associated with Simmel, Cooley, Mead and the Chicago School of symbolic interactionism. Beginning in the 1960s and 1970s, symbolic communication has become central to a remarkably wide range of theories. These theorists have taken different positions on the relationship between interaction, individual and structure; what unifies them is that, in contrast to structure paradigm theorists – who believed that interaction was epiphenomenal – they give symbolic communication a prominent role. Versions of the interaction paradigm that became influential in the United States include symbolic interactionism, ethnomethodology and conversation analysis. Interaction paradigm theories in Europe include Bourdieu's notion of *habitus* (1972/1977), Foucault's discussions of *discourse* (1969/1972), and Habermas' theory of *communicative action* (1987). In both Europe and the United States, the interaction paradigm emerged at about the same time, and in both cases as a response to the inherent tensions of the structure paradigm.

The interaction paradigm denies that the social world has an objective, irreducible structure that constrains individuals in interaction. Macro-level concepts such as social structure and culture are considered to be abstractions that do not really exist nor have any causal force over individuals. Social reality can only be ascribed to concrete interactional processes and it can only be studied in terms of participating individuals' interpretations of it (Blumer 1962: 190). Macrostructural forces never operate directly on individuals, but are mediated through their interpretation by those individuals.

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The interaction paradigm was a necessary intellectual development because it enabled researchers to undertake the close empirical study of the interactional processes of social life. These interactional processes had not been studied by the structure paradigm; a static synchronic focus led it to neglect the dynamic contingency of situated discourse. However, in making this antithetical move, the interaction paradigm was left with a problematic orientation toward emergent frames and structures. The interaction paradigm is uncomfortable with the idea that social phenomena might be irreducibly emergent. For example, the interactional frame remains under-theorized because a theory of the frame is necessarily partially collectivist and partially structuralist, and as such any theory of the frame seems, to an interactionist, to suffer from the same problems as the structure paradigm (Sawyer 2005). This stance has made it difficult for the interaction paradigm to study several aspects of social emergence, including how individual participants are constrained by macro-social forces extending far beyond the encounter, and how individual actions collectively result in the emergence of macro-social phenomena.

The interaction paradigm commonly leads to what I call *interactional reductionism*, the position that sociologists need only to study interaction: "macrophenomena are made up of aggregations and repetitions of many similar microevents" (Collins 1981: 988). Most of the symbolic interactionists and conversation analysts were interactional reductionists; they focus primarily on small-group encounters and do not directly address the emergence of the social from complex systems of individuals.

Although interactional reductionism is the dominant form of the interaction paradigm, there are a few hybrid theorists who have attempted to analyze both interaction and structure and their mutual relations (although without much impact on mainstream sociology). The concept of *discourse*, originating in 1970s French discourse analysis, has been central to many hybrid interaction theories (Sawyer 2002a). During the 1970s, French discourse analysts such as Paul Henry and Michel Pêcheux developed Althusser's concept of ideology into a theory of discourse. Pêcheux (1982) explored the relations that discourses have with ideologies; discourses, like ideologies, develop out of clashes with one another, and there is always a political dimension to writing and speech. Pêcheux explored the relationship between discursive formations and *ideological formations*; for Pêcheux, the discursive formation was the key concept that provided the causal link between social structures and individual consciousness: "Individuals are 'interpellated' as speaking-subjects (as subjects of *their* discourse)

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by the discursive formations which represent 'in language' the ideological formations that correspond to them" (1982: 112). Discourse is a level of analysis that intermediates structure and individual. However, Althusserian discourse theory did not explore how structures emerge from interaction; rather, it emphasized that interaction is determined by structure.

Beginning in the 1990s, an effort known as *critical discourse analysis* emerged from British cultural studies, building on these 1970s notions of discourse and ideology (Fairclough 1995). Critical discourse analysis shares with Althusserian discourse analysis the attempt to simultaneously study both interaction and social structure, considering each as autonomous levels of social reality. However, in practice, critical discourse analysis has rarely documented or explained specific cases in which structural phenomena emerge from interaction. In most cases, the empirical studies of critical discourse analysis demonstrate how social structures reproduce themselves through interaction. These studies often demonstrate the mechanisms of class reproduction – a traditional Marxian concern – in spite of (or with the unwitting participation of) the creative agency of individuals. Critical discourse analysis has not proposed a theory of social emergence.

The lack of a theory of social emergence is a consequence of the failure to theorize a mechanism between social structure and interaction. The interaction paradigm has not explained the mechanisms whereby macro-social phenomena causally influence interaction. And, more importantly, the interaction paradigm does not explain how interaction could causally influence social structure; these influences would require a theory of social emergence, focused on explaining how interactional processes result in the emergence of social properties. Without a well-developed account of how interactional processes result in social emergence, interaction paradigm theorists have been unable to convince other sociologists that conversation is central to sociological theory.

- The interaction paradigm rejects the necessity of examining individuals (the realm of psychology) and macro-social structures (the realm of macro-sociology). Thus it seems that no concept of emergence is necessary.
- The interaction paradigm lacks ontological depth; its conception of reality has a narrow scope around the interaction level, and neglects the causally autonomous properties of both structures and individuals.
- The interaction paradigm has not theorized social constraint; in fact, interactional reductionists reject the existence of such constraint.

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In sum, the interaction paradigm has no theory of social emergence – no explanation of how stable structures emerge from the joint collective actions of individuals engaged in social interaction. Accounting for causal relations between structure, interaction and individual requires a theory of the mechanisms of conversation that give rise to social emergence. A social mechanist approach can address the weaknesses of both structure paradigm and interaction paradigm theories. Social mechanism does so by integrating core elements of both of those paradigms – an acknowledgement that social reality is stratified, and an insistence that conversation represents an autonomous ontological level, mediating between individuals and emergent social phenomena.

Conclusion

The social scientists best known for emphasizing the importance of lower-level causal explanation are those who have advocated a close focus on specific cases – anthropologists explaining a specific cultural practice, historians explaining a specific historical event. It remains a risk that a mechanistic approach could provide wonderful explanations of specific token instances, but degenerate into the study of specific cases, thereby losing all generality and thus being of limited scientific value (see Mayntz 2004). After all, the social sciences that have developed the most elaborated mechanistic accounts of token social phenomena, history and anthropology, are both academic disciplines that resist lawful generalizations.

Even if lower-level causal explanation is limited in generalizability, sociologists could still develop more general scientific explanations in terms of systems and mechanisms, but the description of the system and mechanism would have to incorporate the terms and properties of macro-sociology in addition to individual properties and relations. Although "mechanism" is commonly associated with methodological individualism – because its advocates assume that a social mechanism must be described in terms of an individual's intentional states and relations (e.g. Elster 1998) – there is no reason why science cannot include systems and mechanisms at higher levels of analysis (Wight 2004). After all, individual properties such as intentional states are themselves realized in the lower-level substrate of neurons and their synaptic connections. Non-reductive individualism takes the argument that individual properties should be allowed in scientific explanation, and develops a similar argument to defend social properties and social explanation (Sawyer 2002b).

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I have argued that conversational groups meet many of the criteria for emergent complex systems. However, a sustained argument that improvisational performances are complex systems requires a more extended presentation, and this is not the place for such an argument (although see Sawyer 2003b). Ultimately, the success or failure of micro-explanation in terms of individuals must be determined through empirical research. The social mechanism approach is valuable exactly because it provides a framework that allows us to carry out this exploration.

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